Assignment_VOLUME & SURFACE AREA

1. A cuboid is 15m $\times$12m$\times$10m. Find its length of diagonal, surface area and volume. **Ans :21.49m, 900$\mathbf{m^2}$, 1800$\mathbf{m^3}$**

2. A cuboid is 10m $\times$12m$\times$14m. Find its length of diagonal, surface area and volume. **Ans: 20.97 m, 856$\mathbf{m^2}$, 1680$\mathbf{m^3}$**

3. A cube has edge 14 m. Find its length of diagonal, surface area and volume. **Ans: 14$\sqrt{3}$, 1176$\mathbf{m^2}$, 2744 $\mathbf{m^3}$**

4. A cube has edge 16 m. Find its length of diagonal, surface area and volume. **Ans: 16$\sqrt{3}$, 1536$\mathbf{m^2}$, 4096 $\mathbf{m^3}$**

5. The surface area of a cube is 864 sq metres. Find the volume of the cube. **Ans: 1728 $\mathbf{m^3}$**

6. The surface area of a cube is 1014 sq metres. Find the volume of the cube. **Ans: 2197 $\mathbf{m^3}$**

7. A cube has diagonal 8$\sqrt{\mathbf{3}}$metres long. Find its volume. **Ans: 512 $\mathbf{m^3}$**

8. A cube has diagonal 9$\sqrt{\mathbf{3}}$metres long. Find its volume. **Ans: 729 $\mathbf{m^3}$**

9. Find the volume of a cuboid 8 m$\times$ 7 m $\times$ 6 m.

Ans: 336 $\mathbf{m^3}$

10. Find the volume of a cuboid 9 m$\times$ 8 m $\times$ 5 m.

Ans:360 $\mathbf{m^3}$

11. Find the volume of cube whose edge is 8 m. **Ans: 512 $\mathbf{cm^3}$**

12. Find the volume of cube whose edge is 9 m. A**: 729 $\mathbf{cm^3}$**

13. Find the volume, curved surface area and the total surface area of a cylinder with diameter of base 14 cm and height 28 cm. **Ans: 4312 $\mathbf{cm^3}$, 1232 $\mathbf{cm^2}$, 1540 $\mathbf{cm^2}$**

14. Find the volume and surface area of a cuboid 15m long, 16m broad and 4m high. **Ans: 960$\mathbf{m^3}$, 728$\mathbf{m^2}$.**

15. Find the volume and surface area of a cuboid 12m long, 10m broad and 8m high. **Ans: 960$\mathbf{m^3}$, 592$\mathbf{m^2}$.**

16. The internal measurements of a box with lid are 110$\times$ 70 $\times$ 30 cm^3 and the wood of which it is made is 1.5 cm thick. Find the volume of wood. **Ans: 41,217 $\mathbf{cm^3}$**

17. The internal measurements of a box with lid are 120$\times$ 60 $\times$ 25 cm^3 and the wood of which it is made is 2.5 cm thick. Find the volume of wood. **Ans: 63,750 $\mathbf{cm^3}$**

18. The volume of water measured on a rectangular field 500m$\times$300m is 6000m^3. Find the depth (amount) of rain that has fallen. **Ans: 4 cms.**

19. The volume of water measured on a rectangular field 400m$\times$200m is 8000m^3. Find the depth (amount) of rain that has fallen. **Ans: 10 cms.**

20. Three cubes of metal whose edges are 3, 4 and 5cms. Respectively, are melted and formed into a single cube. Find the edge of the new cube formed. **Ans: 6 cms.**

Assignment_VOLUME & SURFACE AREA

21. Three cubes of metal whose edges are 1, 6 and 8cms. Respectively, are melted and formed into a single cube. Find the edge of the new cube formed. **Ans: 9 cms.**

22. Two cubes have their volumes in the ratio 27:64. Find the ratio of their surface areas.
Ans: 9:16

23. Two cubes have their volumes in the ratio 125:216. Find the ratio of their surface areas.
Ans: 25:36

24. If the capacity of cylindrical tank is 4224 $\mathrm{m}^3$and the diameter of its base is 16m, then find the depth of the tank. **Ans: 21 m.**

25. If the capacity of cylindrical tank is 7128 $\mathrm{m}^3$and the diameter of its base is 12m, then find the depth of the tank. **Ans: 63 m.**

26. The radii of two cylinders are in the ratio 2:7 and their heights are in the ratio of 3:4. Find the ratio of their curved surface areas. **Ans: 3:14**

27. The radii of two cylinders are in the ratio 10:7 and their heights are in the ratio of 3:5. Find the ratio of their curved surface areas. **Ans: 6:7**

28. Find the slant height, volume, curved surface area and the whole surface area of a cone of radius 7 cm and height 24 cm. **Ans: 25m, 1,232$\mathbf{m^3}$,550 $\mathbf{cm^2}$,704 $\mathbf{cm^2}$**

29. Find the length of canvas 2.31m wide required to build a conical tent of base radius 21 metres and height 28 metres. **Ans: 1000m.**

30. Find the length of the longest pole that can be placed in a room 12 m long, 3m broad and 4m high. **Ans: 13 m.**

31. Find the length of the longest pole that can be placed in a room 15m long, 18m broad and 26m high. **Ans: 35 m.**

32. Find the cost of canvas 1.75 m wide at Rs. 50 per metre required to build a conical tent of base radius 21 metres and height 28 metres. **Ans: Rs. 66,000 m.**

33. Find the number of bricks, each measuring 24cm$\times$ $12\mathrm{cm} \times 8\mathrm{cm}$, required to construct a wall 24m long, 8m high and 60 cm thick, if 10% of the wall is filled with mortar? **Ans: 45,000.**

34. Three solid cubes of sides 1cm, 6cm and 8cm are melted to form a new cube. Find the surface area of the cube so formed. **Ans: 486 $\mathbf{cm^2}$**

35. A cylinder of length 3 meter and diameter 30 cms is melted down and cast into spheres of radius 5 cms. How many spheres can be made? **Ans: 405**

36. A well with 20 m inside diameter is dug 10 m deep. Soil taken out of it has been evenly spread all around it to a width of 30 m to form an embankment. Find the height of the embankment**. A: $\frac{2}{3}$ m**

37. A well with 14 m inside diameter is dug 8 m deep. Soil taken out of it has been evenly spread all around it to a width of 28 m to form an embankment. Find the height of the embankment. **A: $\frac{1}{3}$ m**

38. A cylindrical tub of radius 21 cm contains water upto a depth of 20 cm. a spherical iron ball is drop into the tub and thus the level of water is raised by 10.2 cm. find the radius of the ball? **Ans: 15cm approx.**

Assignment_VOLUME & SURFACE AREA

39. A cylindrical tub of radius 28 cm contains water upto a depth of 20 cm. a spherical iron ball is drop into the tub and thus the level of water is raised by 15.75 cm. find the radius of the ball? **Ans: 21cm**.

40. A cone of height 14 cms and base diameter 12 cms is carved from a rectangular block of wood 10 cms $\times$ 8cms $\times$ 7cms. Find the approx percentage of wood wasted. **Ans: 6%.**